One write, three readers

Every agent leaves a trail. Only your team can read it, and anyone can read the code.

THE PROBLEM

Teams run several coding agents — Claude Code, Codex, OpenCode, Goose — some on open-weight models they host. Each keeps a partial record, and none controls how its agents reach the world outside.

Sessions scatter. The CLI clears them at 30 days; each vendor keeps only its own.

Edits escape. Shell and subagent edits skip checkpoints.

Handoffs break. Work rarely follows across harnesses.

Approvals don't travel. Each vendor's, unsigned and unbound to the action.

Reach is shared, not scoped. One grant covers every session and run.

Local runs go unrecorded. No vendor keeps a record you control.

FOR DEVELOPERS

Never lose a session or an edit, whichever harness made it.

Every session goes to a vault under the developer's own keys, and every change gets a restore point, whatever tool path or model made it.

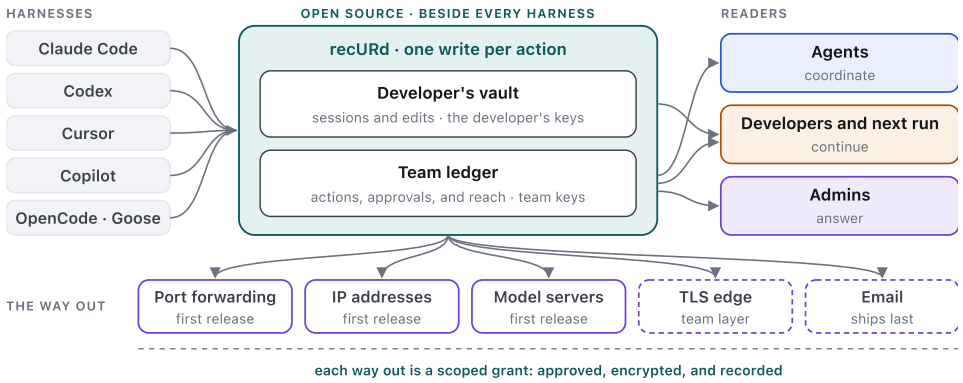
FOR SECURITY BUYERS

Every way your agents reach the outside world is private, approved, and on a record only your team can read.

Forwards, private addresses, and model servers first, then public names, then mail. Each is a scoped grant with a signed receipt; every run lists what it could not see.

HOW IT WORKS

recURd runs a small open-source sidecar beside any harness. It captures each action once — from the harness's hooks, the filesystem, and its own traffic — and writes it once: the raw session to the developer's vault, a signed entry to the team's ledger. Agents coordinate, developers continue, admins answer.



WHY ONE WRITE MATTERS

Audit-only logs are overhead, so they get bypassed. Here agents need the entries to coordinate and developers need them to recover, so the record fills itself.

OPEN SOURCE

You don't have to trust us. recURd, its relay, and the entry format are open; self-host any part, air-gapped if needed. It is plain git underneath.

Private from us. Actions and approvals complete for your admins. Session contents stay with their author unless shared; in teams, run titles are visible by default.

Never lose a session or an edit

Whichever harness made it. This is why a developer installs recURd.

~280
developers report agent changes they could not review or undo.

~200
report lost or hidden sessions. The Claude Code CLI clears local transcripts after 30 days by default.
Claude Code #62476

139
report handoffs that break between runs, machines, or harnesses.

66%
of developers switched AI tools in the past year, so the record must outlive any one harness.
Stack Overflow survey 2026

The first three are counts of distinct people among 914 developer voices on GitHub, Hacker News, and forums, October 2026.

Session vault

WHY THEY INSTALL

Every session from every harness and model, under the developer's own keys. Kept on your retention policy, not a vendor's timer; searchable on the developer's machine; secrets redacted before saving.

Change set and restore

WHY IT STAYS ON

Every file a run changed by any tool path, including shell commands, subagents, and reverts. Restore previews first, stays inside the run's own changes, and is recorded.

Pick up anywhere

Continue a run on another machine, in another harness or model, or as a teammate. Handoff notes record decisions, not transcripts.

Context that loads

Team context written into each harness's own files, such as `AGENTS.md` and `CLAUDE.md`, with a warning before a harness truncates it.
Codex cuts AGENTS.md at 32 KB, with only a log-file warning.

Agents that find each other's work

Claims and leases across harnesses, so two agents don't edit the same files, and the next run starts where the last one stopped.
A hypothesis until the pilots measure it.

Yours unless you share

Admins see actions and approvals, not transcripts. Session contents open only when the developer shares them, and every admin read is visible to the developer. In teams, run titles are visible to teammates by default.

A DAY WITHOUT LOSING ANYTHING

TIME	WHAT HAPPENS	WHAT RECURD KEEPS
09:10	A Claude Code session runs on a laptop.	The full session, in the developer's vault
10:20	A shell command edits files that the harness's checkpoints miss.	Every changed file, in the run's change set
11:05	The run goes wrong.	A preview, then a restore of only that run's changes
13:30	The developer continues in OpenCode, on a local model, on another machine.	The handoff note and the session, ready to resume
16:00	A teammate's agent starts on the same module.	A live claim, so that agent picks other work

Free for one person
Any number of agents and harnesses: the vault, restore, handoffs, and approval history.

Nothing to change
Captures from each harness's own hooks, deployed once per machine, plus a CLI. No new editor.

Open source
Read exactly what it captures and redacts, or build it yourself.

Plain git underneath
Back up the vault, host it yourself, or leave with a clone.

Govern every way out

Every way your agents reach the outside world is private, approved, and on a record only your team can read.

Port forwarding

FIRST RELEASE

A cloud agent reaches a private service or a developer's own machine, without opening the network.

GRANT One agent, one port, one time window, approved by a person.

ENCRYPTED Between the two sidecars. The relay carries only ciphertext.

RECORDS `route` `approval` — `ur bridge forward`

IP addresses

FIRST RELEASE

A permanent private address for each agent and service, plus subnets you choose to route.

GRANT With a policy, agents reach only granted ranges, and leave only through exits you run.

ENCRYPTED Packet by packet, between your sidecars.

RECORDS `route` — `ur tun ip`

TLS edge

TEAM LAYER

A public HTTPS name for a service on a developer's machine, such as a client preview or a webhook.

GRANT Explicit, authenticated by default, and short-lived. Agent control panels are refused.

ENCRYPTED TLS passthrough. The certificate key stays in the sidecar.

RECORDS `publish` — `ur bridge public-tls`

Email

SHIPS LAST

Addresses on your own domain for agents and people, including approval by reply.

GRANT Outside mail is untrusted. First messages to new recipients wait for approval.

ENCRYPTED End to end between members. Outside mail is encrypted on arrival.

RECORDS `ask` `answer` `approval` — `ur id`

REACH IS SHARED, NOT SCOPED, TODAY

Codex cloud supports only Tailscale, which warns Codex "could reach nearly everything in your tailnet." No vendor scopes a hosted run to one service on a developer's machine, with an approver and expiry.

EASY EXPOSURE HAS ALREADY GONE WRONG

Knostic found 1,862 MCP servers open to the internet; none of 119 checked required auth to list tools. And 68% of organizations cannot tell agent activity from human (CSA, Aembit-sponsored).

WHAT ADMINS CAN ANSWER

- Which agents, harnesses, and models touched which repos, for which sponsor.
- Who approved each action: one action ID, SSH-signed for high-risk steps.
- Proof for a client that a person approved a change, without the code.
- One switch revokes an agent's forwards, addresses, names, and mail.

SAFE BY DEFAULT

recURd holds the three legs of the lethal trifecta — private data, untrusted input, the way out — apart.

- Untrusted mail cuts off send, forward, and publish until a person approves.
- Edits to `CLAUDE.md`, hooks, and `.mcp.json` go to review.
- Shared writes are scanned for secrets, and recURd keeps no long-lived ones.

WHAT ENCRYPTION CHANGES, AND WHAT IT DOESN'T

With customer-managed keys the vendor still decrypts and may keep records outside your key. recURd's relay and storage hold no key, so a subpoena or breach yields only ciphertext and the metadata listed here; blindness shortens the review but a DPA, subprocessor list, and breach notice remain. **Where it ends:** a hosted model sees what the harness sends unless you run it; outside mail is ordinary SMTP; signed membership stops a breached server adding a reader.

Open all the way down

Open source, so no one has to take our word for it. Run the models yourself, and prompts and code never leave your control.

WHAT CHANGES WHEN THE MODEL IS YOURS

	HOSTED MODEL	OPEN WEIGHTS YOU RUN
Who sees prompts and code	The model provider	Only your team
Vendor record of agent work	Each vendor's, for its own product; GitLab now ingests some	None, so the record is the one you keep
What URnetwork holds	Ciphertext	Ciphertext, or nothing if you self-host
What the record can prove	Which harness and which model name	Also the weight digest, quantization, and engine

Anthropic's Compliance API and OpenAI's compliance logs cover their own products. Open-weight teams get the whole privacy chain, and the most need for a record.

Model access

FIRST RELEASE

Agents reach the inference servers you run, such as vLLM, SGLang, Ollama, or llama.cpp, and nothing else.

GRANT	Per agent and per model, approved by a person, revoked with the agent.
ENCRYPTED	Between the sidecars. No vendor sits in the path.
SIGNS	Per request: the weight digest, quantization, and engine, each labelled by evidence level.

RECORDS `route` `model` `ur bridge forward`

WHO IT'S FOR

Mixed fleets — teams that run some agents on open-weight models

The common case, not a niche: ~11–20% of agent-model users name an open-weight family and ~2% name nothing else (Stack Overflow 2026); on OpenRouter open weights carry ~80% of coding tokens but ~33% of spend.

Developers and homelabs

Adoption and credibility, but little revenue. Open tools draw big followings: OpenCode has 224k GitHub stars, Ollama 186k, llama.cpp 133k. Stars measure attention, not use.

WHERE OPEN TOOLS ALREADY STAND

PIECE	OPEN TOOLS TODAY	WHERE WE STAND
Reach	Free to self-host: frp (110k stars), Headscale (45k), zrok, and iroh. zrok's hosted service runs "the same code base" people self-host.	Free in ours too. We add per-agent grants and receipts.
Agents and shared models	Block's Buzz: a signed, self-hostable relay that runs Goose, Codex, and Claude Code over ACP, streaming every event to one log.	Partner first. We record Goose like any harness.
Encrypted git	Radicle calls its private repos "unfit for use today"; git-crypt cannot revoke access; git-remote-gcrypt syncs only on push and pull.	Ours: real time, with team key rotation.
A signed record of agent work	About two dozen competing IETF drafts and an AUDIT BoF (17 Nov); OCSF 1.9 defines signed agent events. Agent Receipts and nono already sign tool calls and sessions.	Ours: one record across harnesses binding approvals, edits, and reach under your keys.

OPEN CORE, HOSTED BLIND

Free and open: recURd, the admin app, relay, and entry format, and self-hosting any part, air-gapped if needed. **Paid:** relays, storage, and an edge we run and cannot read; names, certificates, and mail delivery; the team layer; support for your own installs. iroh, zrok, and Tailscale earn the same way.

recURd is in design. GitHub stars fetched 1 November 2026.

Sources: sidecar/research_notes, GitHub, Stack Overflow 2026

What's ours, and who pays

The tools are table stakes. The record under your keys is the product.

WHERE WE STAND, PIECE BY PIECE

PIECE	WHO ALREADY SELLS IT	WHERE WE STAND
Reach: forwards, private addresses, public names	ngrok, Cloudflare Tunnel, and Tailscale, which is end-to-end encrypted and Codex cloud's only VPN	At par, and free when self-hosted. We add per-agent grants and receipts.
Agent mail	AgentMail, Resend Inboxes, Cloudflare Email Service	At par, so it ships last.
Coordination across agents	Claude Code agent teams, Codex message boards, Block's Buzz, Warp, and Entire	Also span harnesses; we don't lead. Claims stay a free hypothesis.
A signed team ledger across harnesses, under customer keys	Happy and Infisical Agent Vault keep host-unreadable records; none binds approvals, edits, and reach in one.	Ours.
Sessions and actions under separate keys	Span keeps sessions private by access control, not keys.	Ours.

Why the gap stays open: harness owners earn when work stays on their runtime, and gateway and recorder vendors are built on reading the data. That is an inference about incentives. Anthropic, OpenAI, GitLab, and Block moved furthest this year; watch them.

WHO PAYS

Solo **Free, open source**	One person, any number of agents: the vault, restore, handoffs, forwards, and model access.
Team **$20–40 / seat***	From the second person, hosted: the signed ledger, cross-harness attribution, per-client cost and policy, reach receipts, SSO/SCIM, exports.
Assured **Annual**	For regulated and defense buyers: change records, a self-hosted licence and support, compliance mappings, an air-gapped install; on your servers.

*Team dev tools run $20–50 a seat; SSO plans start near $500/mo. Agents are never metered; storage, edge traffic, and mail are. Self-hosting is free.

WHERE WE START

Teams of 5 to 150 engineers on two or more harnesses, where a contract already demands proof:

- Agencies and outsourcers under client AI terms, one AI policy per client
- Suppliers to banks and defense primes with flow-down AI terms (Citi, Lockheed)
- Teams whose cloud agents need private reach
- Fintech and other regulated teams, including mixed open-weight fleets

MONEY IS MOVING INTO THIS SPACE

- **Entire:** $60M seed (Feb 2026), agent sessions in git
- **AgentMail:** $6M seed (Mar 2026), agent email
- **Tailscale:** $160M Series C (Apr 2025), now for agents
- **Span:** $25M (Nov 2025), agent traces
- **Keycard** ($38M), **Mem0** ($24M): identity, memory
- **Agent 365:** $15 per user, agents unlicensed

Now
Questionnaires ask which AI a team uses and whether a human reviews its work.

From Dec 2027
If finalized, SOC 2 engagements use criteria whose AI points of focus prompt auditors' questions; they require no specific control.

The moat is service, not lock-in
The format is open and portable, so nothing traps the record. Teams stay because hosting, support, and coverage are good.

Proof before scale

One open-source capture engine, ten to fifteen design partners, and published trust documents come before the rest of the build.

WHAT SHIPS, IN ORDER

First release, open source

The vault, change sets and restore, approval receipts, run inventory and agent identity, handoffs and claims, commit-to-run links, and the instruction-file writer.

Reach: port forwarding, private IP addresses (Linux and WSL2 first), and access to model servers you run.

Next: the team layer

The shared signed ledger, SSO and SCIM, SIEM export, retention, the verifiable log, client packets, reach receipts, and a supported self-hosted install.

Reach: the TLS edge, and approval by email reply.

Last: inbox and assurance

The visible audit member, key-escrow options, legal hold, compliance mappings, and an air-gapped install.

Reach: mail with the outside world on your domain, and group inboxes.

Each stage ships whole. Reach comes first, built on code URnetwork already runs; email, the heaviest build, comes last.

TEN TO FIFTEEN DESIGN PARTNERS

Teams of 5 to 150 engineers on two or more harnesses, drawn from the beachhead, including two or three that run open weights. Each question has a measure, and a decision if the answer is no.

QUESTION	WHAT WE MEASURE	IF THE ANSWER IS NO
Do developers keep it running?	Daily use of the vault and restore, per developer	Rework the developer hook before building the team layer
Does the ledger sell?	Conversion to Team when a second person joins	Move the free line or the Team price
Does coordination earn its place?	Duplicate work and conflicting edits, with and without claims	Keep claims as a free feature, not a selling point
Is reach a wedge?	Cloud-agent grants approved and used each week	Keep reach as a feature, and lead with the record
Will teams pay?	Paid conversions at $20–40 a seat	Re-price before launch
Do mixed-fleet teams adopt it, and pay?	Model calls recorded through recURd; hosted versus self-hosted relays	Sell the team layer and support, not hosting

TRUST DOCUMENTS FIRST

Before "end-to-end encrypted" appears in public:

- recURd, the admin app, relay, and entry format ship open source, with signed releases.
- The security page discloses that the cryptography is not yet independently audited.
- A threat model assumes a malicious server and lists all metadata our servers see.
- A data-flow diagram and subprocessor list, a self-service DPA, and a dated warrant canary.
- A plain statement: recURd has no token, and uses URnetwork's network only for worldwide reach.

RISKS WE MANAGE

- **Scope.** One capture engine and one entry format; each stage ships whole.
- **Completeness.** Hook behavior differs by harness (some fail closed, some open), so each run states what the record could not see.
- **Free built-ins and self-hosting.** We sell hosting we can't read, the team layer, and support.
- **Open rivals.** Block's Buzz is closest; we partner first and record Goose like any harness.
- **Reputation.** Trust documents come before claims, and agents leave only through exits you run.